

University of Eswatini
Practices in Software Engineering - CSC392

Software Requirements Specification (SRS)
Smart Attendance System

Structured according to IEEE Std 830-1998, using Annex A.7: Template of SRS Section 3 organized by functional hierarchy.

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- Version 1.0 (11 May 2026): Initial SRS structured using IEEE Std 830-1998 Annex A.7 functional hierarchy. Prepared by Project Team.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Smart Attendance System. It is intended for lecturers, students, the development team, testers, and project evaluators. The document provides a clear baseline for system design, implementation, testing, and validation.

1.2 Scope

The Smart Attendance System is a software-hardware integrated solution for automating student attendance recording using barcode scanning technology. The system will allow lecturers to authenticate, create attendance sessions, scan student ID card barcodes, record attendance, and generate attendance reports. The system will reduce manual attendance errors, improve data integrity, and support real-time tracking and reporting.

The system does not include automatic biometric identification in this version. A webcam may be used as an alternative barcode scanning input where a physical scanner is unavailable.

1.3 Definitions, Acronyms, and Abbreviations

- **SRS:** Software Requirements Specification.
- **SDD:** Software Design Description.
- **API:** Application Programming Interface.
- **DBMS:** Database Management System.
- **Barcode:** Machine-readable representation of a student identifier printed on a student ID card.
- **Attendance Session:** A scheduled class attendance event opened by a lecturer for a specific course, class, date, and time.
- **Registered Student:** A student enrolled or assigned to a course or class list.
- **Unregistered Student:** A student whose ID is valid in the student database but who is not registered for the selected course.

1.4 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- IEEE Std 1016-2009, IEEE Standard for Information Technology - Systems Design - Software Design Descriptions, referenced for later design documentation alignment.
- Project Plan: Smart Attendance System, University of Eswatini, Practices in Software Engineering - CSC392.

1.5 Overview

Section 1 introduces the purpose, scope, terms, and references for this SRS. Section 2 describes the product perspective, users, assumptions, constraints, and major functions. Section 3 gives the specific requirements and follows the IEEE Std 830-1998 Annex A.7

functional hierarchy structure: external interfaces, information flows, process descriptions, performance requirements, design constraints, software quality attributes, and other requirements.

2. Overall Description

2.1 Product Perspective

The Smart Attendance System is a web-based application supported by attendance scanning hardware. The frontend provides the lecturer and administrative user interface. The backend exposes application services and business rules. The database stores lecturers, students, courses, class schedules, attendance sessions, attendance records, and reports. The barcode scanner or webcam provides student identification input.

The system operates as part of an academic environment where student ID cards, class schedules, and course data are available. It is intended to replace or reduce paper-based attendance registers.

2.2 Product Functions

- **Authentication:** Allow lecturers and administrators to log in securely before accessing attendance functions.
- **Course and Schedule Access:** Allow lecturers to view assigned courses, classes, and scheduled attendance sessions.
- **Attendance Session Creation:** Allow lecturers to open an attendance session for a selected course and class period.
- **Barcode Scanning:** Capture a student ID barcode using a scanner or webcam alternative.
- **Student Validation:** Check whether the scanned student exists and whether the student is registered for the selected course.
- **Attendance Recording:** Store valid attendance records and prevent duplicate attendance entries for the same session.
- **Exception Handling:** Flag duplicate, invalid, or unregistered student scans for lecturer review.
- **Reporting:** Generate attendance reports by course, session, date range, or student.

2.3 User Characteristics

- **Lecturer:** Academic user with basic computer literacy and knowledge of course attendance rules. Main system use: Creates attendance sessions, scans students, views and exports reports.
- **Student:** Learner with a student ID card containing a barcode. Main system use: Presents student ID card for scanning and may view personal attendance where supported.
- **Administrator:** Technical or departmental user with higher access privileges. Main system use: Manages users, courses, students, schedules, and data corrections.

2.4 Constraints

- The system shall support barcode scanner input and a webcam alternative where a scanner is unavailable.
- The system shall use a web browser interface for normal operation.
- The planned technology stack is React for the frontend, Node.js with Express for the backend, and PostgreSQL for persistent data storage.
- The system shall protect attendance data from unauthorized access by applying role-based access control.
- The system shall support validation checks to reduce incorrect scans and duplicate records.

2.5 Assumptions and Dependencies

- Student ID cards contain readable and unique barcode values.
- Course, student, and class schedule data will be available before attendance sessions are created.
- Lecturers have access to a supported browser and a working scanner or webcam.
- Network connectivity is available between the frontend, backend, and database during normal operation.
- The project team will follow iterative development and testing to reduce integration risks.

2.6 Apportioning of Requirements

- **Version 1.0:** Lecturer authentication, attendance session creation, barcode scanning, attendance recording, duplicate prevention, reporting.
- **Future Version:** Mobile student attendance view, email notifications, facial recognition, advanced analytics, integration with a university student information system.

3. Specific Requirements

This section follows the IEEE Std 830-1998 Annex A.7 structure for requirements organized by functional hierarchy.

3.1 External Interface Requirements

3.1.1 User Interfaces

- **UI-01:** The system shall provide a login page for lecturers and administrators. Priority: Essential. Verification: Inspection/Test.
- **UI-02:** The system shall provide a lecturer dashboard showing assigned courses and available attendance actions. Priority: Essential. Verification: Inspection/Test.
- **UI-03:** The system shall provide an attendance session page with scan input, student identification result, and attendance status messages. Priority: Essential. Verification: Inspection/Test.
- **UI-04:** The system shall display clear messages for successful scans, duplicate scans, invalid barcode values, and unregistered students. Priority: Essential. Verification: Test.

- **UI-05:** The system shall provide report screens for filtering attendance by course, session, date range, and student. Priority: Essential. Verification: Inspection/Test.

3.1.2 Hardware Interfaces

- **HI-01:** The system shall accept student barcode input from a barcode scanner connected to the lecturer computer. Priority: Essential. Verification: Test.
- **HI-02:** The system shall support a webcam or phone/webcam alternative for barcode capture where the scanner is unavailable. Priority: Conditional. Verification: Test.
- **HI-03:** The system shall not require special student hardware beyond an ID card with a readable barcode. Priority: Essential. Verification: Inspection.

3.1.3 Software Interfaces

- **SI-01:** The frontend shall communicate with the backend through defined API endpoints for authentication, courses, sessions, scans, records, and reports. Priority: Essential. Verification: Inspection/Test.
- **SI-02:** The backend shall store and retrieve persistent data from a PostgreSQL database. Priority: Essential. Verification: Inspection/Test.
- **SI-03:** The system shall support modern web browsers used on student or lecturer laptops. Priority: Essential. Verification: Test.

3.1.4 Communications Interfaces

- **CI-01:** The system shall use HTTP or HTTPS communication between the browser frontend and the backend server. Priority: Essential. Verification: Inspection/Test.
- **CI-02:** The system shall return structured responses for successful operations and error conditions. Priority: Essential. Verification: Test.
- **CI-03:** Where deployed on a network, the system shall protect authenticated communication using secure transport when available. Priority: Conditional. Verification: Inspection/Test.

3.2 Functional Requirements

3.2.1 Information Flows

3.2.1.1 Data Flow Diagram 1: Lecturer Authentication and Session Setup

- **Data Entities:** Lecturer, Login Credentials, Course, Class Schedule, Attendance Session.
- **Pertinent Processes:** Authenticate lecturer; load assigned courses; select course and schedule; create attendance session.

- **Topology:** Lecturer -> Web Interface -> Backend API -> PostgreSQL Database -> Web Interface.

3.2.1.2 Data Flow Diagram 2: Student Identification and Attendance Recording

- **Data Entities:** Student, Barcode Value, Course Registration, Attendance Session, Attendance Record, Exception Record.
- **Pertinent Processes:** Capture barcode; decode ID; validate student; check course registration; prevent duplicate attendance; store attendance record; display scan result.
- **Topology:** Barcode Scanner/Webcam -> Web Interface -> Backend API -> PostgreSQL Database -> Web Interface.

3.2.1.3 Data Flow Diagram 3: Attendance Reporting

- **Data Entities:** Course, Student, Attendance Session, Attendance Record, Report Filter, Report Output.
- **Pertinent Processes:** Select report filters; retrieve attendance data; calculate totals and percentages; display or export report.
- **Topology:** Lecturer/Admin -> Report Interface -> Backend API -> PostgreSQL Database -> Report Output.

3.2.2 Process Descriptions

3.2.2.1 Process 1: Authenticate Lecturer

- **Input Data Entities:** Username/email and password.
 - **Algorithm or Formula of Process:** Receive credentials, validate required fields, compare credentials with stored user record, create authenticated session or return error.
 - **Affected Data Entities:** Lecturer account, authentication session/token, optional audit log.
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- **FR-01:** The system shall authenticate lecturers before allowing access to attendance functions. Priority: Essential. Verification: Test.
 - **FR-02:** The system shall reject invalid login credentials and display an appropriate error message. Priority: Essential. Verification: Test.

3.2.2.2 Process 2: Manage Course and Schedule Data

- **Input Data Entities:** Course details, class schedule details, lecturer assignment.

- **Algorithm or Formula of Process:** Store, update, and retrieve course and schedule data needed to create attendance sessions.
 - **Affected Data Entities:** Course, class schedule, lecturer-course assignment.
-
- **FR-03:** The system shall allow authorized users to maintain course, student, and class schedule information. Priority: Essential. Verification: Inspection/Test.
 - **FR-04:** The system shall display only relevant courses and schedules to the authenticated lecturer. Priority: Essential. Verification: Test.

3.2.2.3 Process 3: Create Attendance Session

- **Input Data Entities:** Lecturer, course, class schedule, date, start time, optional session notes.
 - **Algorithm or Formula of Process:** Validate lecturer access, verify selected course and schedule, create a session with status Open, and record session start time.
 - **Affected Data Entities:** Attendance session.
-
- **FR-05:** The system shall allow a lecturer to create an attendance session for a selected course and class period. Priority: Essential. Verification: Test.
 - **FR-06:** The system shall prevent creation of duplicate open sessions for the same course and class period. Priority: Essential. Verification: Test.

3.2.2.4 Process 4: Capture Barcode and Identify Student

- **Input Data Entities:** Barcode value from scanner or webcam.
 - **Algorithm or Formula of Process:** Capture barcode input, sanitize the barcode value, search for matching student identifier, and return student details if found.
 - **Affected Data Entities:** Student, scan event.
-
- **FR-07:** The system shall capture a barcode value when a student ID card is scanned. Priority: Essential. Verification: Test.
 - **FR-08:** The system shall match the barcode value to exactly one student record where possible. Priority: Essential. Verification: Test.

3.2.2.5 Process 5: Validate Registration and Duplicate Attendance

- **Input Data Entities:** Student, course registration, attendance session, existing attendance records.
 - **Algorithm or Formula of Process:** Check whether the student is registered for the course and whether the student already has an attendance record for the open session.
 - **Affected Data Entities:** Course registration, attendance record, exception record.
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- **FR-09:** The system shall verify that a scanned student is registered for the selected course. Priority: Essential. Verification: Test.
 - **FR-10:** The system shall prevent more than one attendance record for the same student in the same attendance session. Priority: Essential. Verification: Test.

3.2.2.6 Process 6: Record Attendance

- **Input Data Entities:** Validated student, open attendance session, timestamp.
 - **Algorithm or Formula of Process:** Create attendance record with student ID, session ID, timestamp, status Present, and capture method.
 - **Affected Data Entities:** Attendance record.
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- **FR-11:** The system shall record attendance automatically after a valid scan. Priority: Essential. Verification: Test.
 - **FR-12:** The system shall record the date and time of each attendance entry. Priority: Essential. Verification: Test.

3.2.2.7 Process 7: Handle Invalid, Duplicate, or Unregistered Scans

- **Input Data Entities:** Scan result, student registration status, existing attendance record status.
 - **Algorithm or Formula of Process:** Classify scan result as invalid, duplicate, unregistered, or valid; display message; store exception where review is needed.
 - **Affected Data Entities:** Exception record, scan log, attendance session.
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- **FR-13:** The system shall display a clear error message when the barcode does not match any student. Priority: Essential. Verification: Test.
 - **FR-14:** The system shall flag unregistered students instead of silently accepting them as normal attendance records. Priority: Essential. Verification: Test.

- **FR-15:** The system shall display a duplicate warning when a student has already been marked present for the session. Priority: Essential. Verification: Test.

3.2.2.8 Process 8: Close Attendance Session

- **Input Data Entities:** Open attendance session and lecturer confirmation.
 - **Algorithm or Formula of Process:** Validate lecturer ownership or authorization, set session status to Closed, and prevent further attendance records unless reopened by an authorized user.
 - **Affected Data Entities:** Attendance session, attendance records.
-
- **FR-16:** The system shall allow the lecturer to close an open attendance session. Priority: Essential. Verification: Test.
 - **FR-17:** The system shall prevent normal attendance recording into a closed session. Priority: Essential. Verification: Test.

3.2.2.9 Process 9: Generate Attendance Reports

- **Input Data Entities:** Report filter criteria, course, session, student, date range, attendance records.
 - **Algorithm or Formula of Process:** Retrieve matching attendance records, calculate totals and attendance percentages, group records by selected criteria, and present output.
 - **Affected Data Entities:** Report output, attendance records.
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- **FR-18:** The system shall generate attendance reports by course, session, student, and date range. Priority: Essential. Verification: Test.
 - **FR-19:** The system shall calculate attendance totals and percentages for selected reports. Priority: Conditional. Verification: Test.
 - **FR-20:** The system shall allow reports to be viewed on screen and exported where supported. Priority: Conditional. Verification: Inspection/Test.

3.2.2.10 Process 10: Manage Users and Roles

- **Input Data Entities:** User account, role, permission, status.
- **Algorithm or Formula of Process:** Authorized administrator creates or updates user accounts and assigns roles controlling system access.
- **Affected Data Entities:** User account, role permission, audit log.

- **FR-21:** The system shall restrict administrative functions to authorized users only. Priority: Essential. Verification: Test.
- **FR-22:** The system shall support at least Lecturer and Administrator roles. Priority: Essential. Verification: Inspection/Test.

3.3 Performance Requirements

- **PR-01:** The system shall process a valid barcode scan and display the attendance result within 2 seconds under normal local network conditions. Priority: Essential. Verification: Performance Test.
- **PR-02:** The system shall load common lecturer dashboard pages within 3 seconds under normal operating conditions. Priority: Conditional. Verification: Performance Test.
- **PR-03:** The system shall support at least 50 students being scanned in one attendance session without data loss. Priority: Essential. Verification: System Test.
- **PR-04:** The system shall support at least 5 concurrent lecturer users during testing deployment. Priority: Conditional. Verification: Load Test.
- **PR-05:** The system shall generate a report for a single course within 5 seconds for a semester-sized data set. Priority: Conditional. Verification: Performance Test.

3.4 Design Constraints

- **DC-01:** The frontend shall be implemented using React unless the project supervisor approves an alternative. Priority: Essential. Verification: Inspection.
- **DC-02:** The backend shall be implemented using Node.js and Express unless the project supervisor approves an alternative. Priority: Essential. Verification: Inspection.
- **DC-03:** The database shall be PostgreSQL unless the project supervisor approves an alternative. Priority: Essential. Verification: Inspection.
- **DC-04:** The system shall use modular implementation for authentication, attendance, and reporting components. Priority: Essential. Verification: Inspection.
- **DC-05:** The system shall maintain database integrity using appropriate primary keys, foreign keys, and uniqueness constraints for attendance records. Priority: Essential. Verification: Inspection/Test.

3.5 Software System Attributes

3.5.1 Reliability

- **SA-R-01:** The system shall preserve stored attendance records after application restart. Priority: Essential. Verification: Recovery Test.

- **SA-R-02:** The system shall reject incomplete or invalid attendance records before database storage. Priority: Essential. Verification: Test.

3.5.2 Availability

- **SA-A-01:** The system shall be available during scheduled class attendance periods when the server, database, network, and scanning device are operational. Priority: Essential. Verification: System Test.
- **SA-A-02:** The system shall provide clear error messages when the scanner, database, or network is unavailable. Priority: Conditional. Verification: Test.

3.5.3 Security

- **SA-S-01:** The system shall require authentication before allowing access to lecturer or administrative functions. Priority: Essential. Verification: Test.
- **SA-S-02:** The system shall restrict attendance modification and administrative actions by role. Priority: Essential. Verification: Test.
- **SA-S-03:** The system shall not display passwords in plain text. Priority: Essential. Verification: Inspection/Test.
- **SA-S-04:** The system shall keep an audit trail for manual attendance corrections where supported. Priority: Conditional. Verification: Inspection/Test.

3.5.4 Maintainability

- **SA-M-01:** The system source code shall be organized into clearly separated modules for authentication, attendance, scanning, reporting, and data access. Priority: Essential. Verification: Inspection.
- **SA-M-02:** The system shall include documentation for main modules and setup steps. Priority: Conditional. Verification: Inspection.

3.5.5 Portability

- **SA-P-01:** The user interface shall run in modern desktop web browsers used by the project team. Priority: Essential. Verification: Test.
- **SA-P-02:** The system shall support deployment on local development machines for testing and demonstration. Priority: Essential. Verification: Test.

3.5.6 Usability

- **SA-U-01:** A lecturer shall be able to open an attendance session and begin scanning with minimal navigation steps after login. Priority: Essential. Verification: Usability Test.
- **SA-U-02:** The scanning page shall use visible success, warning, and error feedback to reduce lecturer confusion during attendance taking. Priority: Essential. Verification: Usability Test.

3.6 Other Requirements

- **OR-01:** The system shall retain attendance records for the duration of the academic semester unless deleted by an authorized administrator. Priority: Essential. Verification: Inspection/Test.
- **OR-02:** The system shall support backup of the PostgreSQL database through standard database backup procedures. Priority: Conditional. Verification: Inspection.
- **OR-03:** The system shall avoid unnecessary exposure of personal student data in reports. Priority: Essential. Verification: Inspection/Test.
- **OR-04:** The system shall provide enough test data and test cases to support unit, integration, and system testing. Priority: Conditional. Verification: Inspection.

Appendix A: Data Dictionary

Data Entity	Description	Key Attributes
Lecturer	Academic staff user who manages attendance sessions.	lecturer_id, name, email, password_hash, role
Student	Student record identified by a barcode value.	student_id, student_number, name, barcode_value, programme
Course	Academic course for which attendance is recorded.	course_id, course_code, course_name, lecturer_id
Class Schedule	Planned class time for a course.	schedule_id, course_id, day, start_time, end_time, venue
Attendance Session	Opened attendance instance for a course and class time.	session_id, course_id, lecturer_id, date, start_time, end_time, status
Attendance Record	Student attendance entry for a session.	record_id, session_id, student_id, timestamp, status, capture_method
Exception Record	Scan event requiring review.	exception_id, session_id, barcode_value, reason, timestamp
Report	Generated view of attendance information.	report_id or query, filters, generated_by, generated_at

Appendix B: Requirements Traceability Matrix

Requirement Group	Related Project Need	Verification Approach
FR-01 to FR-02	Lecturer authentication.	Login tests and access control tests.
FR-05 to FR-06	Attendance session creation.	Functional tests with valid and duplicate sessions.

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FR-07 to FR-12	Barcode scanning, validation, and attendance storage.	Scanner/webcam tests, database tests, duplicate prevention tests.
FR-13 to FR-15	Data error handling for invalid, duplicate, and unregistered scans.	Negative testing and exception tests.
FR-18 to FR-20	Report generation.	Report filter tests and output review.
PR-01 to PR-05	Fast and responsive system.	Performance and load tests.
SA-S-01 to SA-S-04	Security and data integrity.	Role-based access tests and inspection.